HaltIQ Strands study

TERMINAL PAIRED REPLAY CHECKPOINT

Evidence checkpoint: 2026-10-18T20:46:13.640222-05:00

1. Evidence and completion

This report separates completed computation, incomplete coverage and proposed experiments. It does not claim that elapsed monitoring time establishes a completed six-hour experiment.

TRAIN cleanup: writer_unload_verified=True. An interrupted source retains its original planned denominator; neither omitted questions nor a derived subset can silently relabel it complete.

Last recorded TRAIN control state: intentional_rest; reasons: none recorded. A resource hold preserves completed requests rather than repeating them.

Explicit campaign state: completed. Launching a supervisor is not a model forward.

A separate completed training campaign owns TRAIN caches, checkpoint updates and calibration. Its actual stage captures are verified directly; fresh evaluation does not import or recreate those stages.

Evidence	Declared	Observed	Interpretation
Selected TRAIN source	90	90	completed; partitions {'fit': 60, 'tune': 15, 'calibration': 15}; saved trajectory failures 0
TRAIN physical requests	90	90	pending 0; error events 0
TRAIN missing trajectories	90	0	0 completed-call question(s) lack a saved trajectory; 0 questions never dispatched
Untouched Strands baseline	20 before learning	20	gate verified
Pointer-head training	disjoint fit/tune	60	weight/reload proof verified
Fresh paired confirmation	20	20	terminal matched replay

2. Dataset and architecture

The task is short-answer HotpotQA with all supplied distractor paragraphs, titles and sentence order. Questions, evidence and the current draft are public model inputs; references, supporting facts and correctness labels are evaluator-only. Answer metrics use official-style normalized exact match and token F1.

Official TRAIN and public validation must be disjoint by ID and normalized question. The original 600-question training proposal uses 400 fit / 100 tune / 100 calibration cases; its six fixed 100-row source blocks are not an IID sample of all 90,447 TRAIN questions. A resource-driven TRAIN reduction requires a separately adopted manifest before learning, with original partitions and parent provenance preserved.

Selected TRAIN for this report: 90 planned questions, FIT 60 / TUNE 15 / CALIBRATION 15. This source's generation and native features are separate from historical preparation expense. Its answer-blind reduction ranked canonical IDs by a seed42 hash within original partition and public question type, retained original order/partitions, and excluded answers,

outcomes, draft availability and context length from selection. The preserved selection manifest still says PROPOSED; actual execution is established separately by the completed owning campaign.

Dataset attribution: HotpotQA (https://hotpotqa.github.io/), CC-BY-SA 4.0; selected source revision 1908d6afbbead072334abe2965f91bd2709910ab. Preserve source attribution and the recorded ID/question exclusion audit when distributing derived data.

Selected training dataset SHA256: 1a789919c3d563999337109a4d5e97d5cb89dcadd042e1c825b66fcba59df1ff. Explicit cohort manifest status: PROPOSED_NOT_ADOPTED; recorded partition counts: {'fit': 60, 'tune': 15, 'calibration': 15}.

The evaluated Strands stop controller scores binary draft correctness and stops at probability >=0.8 or continues a generic revision loop. Strands accepts at most 4,096 fully rendered tokens, without truncation; overflow remains an explicit coverage failure. The writer has a separate 16k context. Ten total drafts and cap-three parity share recorded trajectories. The released 2B encoder and adapter remain frozen; actual supervised fine-tuning updates the released pointer head only. One EM correctness proxy is not three independently labeled rubric targets. LoRA/backbone training and RL are separate future work.

Confirmation is a once-frozen subset of 20, 50 or 100 public validation questions, not an official hidden-test score. Shared Fixed 1/2/3/5/10 controls use the identical observations. The unchanged Vedanshi v0 export is a schema contract, not proof of identical hardware; canonical planned-N summaries retain failures/zero-draft cases that v0 cannot express.

The shared comparison measures all-planned-question answer EM and token F1, first-draft repairs and regressions, approval precision with coverage, and first-decision AUROC, Brier score and NLL. Question-paired bootstrap intervals use 5,000 draws and seed42; counterfactual prefix costs and actual physical work are separate. These metrics are reported only when their result-specific evidence gates are verified.

3. Environment and model identity

The profile preserves one study GPU owner, four CPU threads, background rests and memory admission. A resource pause does not authorize lowering the floor or closing user applications. Package targets are not installation or kernel evidence.

Pinned target: upstream Strands source at 930e6d5de16309be32579f5631a7369a3322140c; released decider at 216d1ccb46a9e9b036eb9187b248db98480f77d4; Qwen3.5-2B-Base revision b1485b2fa6dfa1287294f269f5fb618e03d52d7c. These are intended source identities until actual weight/inference receipts establish use. New 16k/q8 results must not be pooled with historical 32k/f16 development results.

Item	Recorded evidence
Python / platform	3.12.10 / Windows-11-10.0.26200-SP0
Dated hardware inventory	2026-10-05T20:15:04.1099788-05:00; AMD Ryzen 7 9800X3D 8-Core Processor; NVIDIA GeForce RTX 5070 Ti, 16303 MiB
Installed physical RAM capacity	31.06 GiB
Writer identity	qwen3:8b / digest 500a1f067a9f782620b40bee6f7b0c89e17ae61f686b92c24933e4ca4b2b8b41
Writer settings	{"context_tokens": 16384, "output_tokens": 512, "temperature": 0, "think": false, "threads": 4}

Item	Recorded evidence
Serving profile	{"LLAMA_ARG_CACHE_RAM": "0", "OLLAMA_FLASH_ATTENTION": "true", "OLLAMA_GPU_OVERHEAD": "3221225472", "OLLAMA_KV_CACHE_TYPE": "q8_0", "OLLAMA_MAX_LOADED_MODELS": "1", "OLLAMA_NUM_PARALLEL": "1"}
Resource admission	{"minimum_idle": 5, "ram_gib": 7, "rest_ratio": 1, "threads": 4, "vram_gib": 3}
Installation / health	completed; CUDA health present
Actually installed packages	{"accelerate": "1.15.0", "numpy": "2.5.4", "peft": "0.21.0", "torch": "2.7.1+cu128", "transformers": "5.17.0"}
Download receipts / target packages	53/53 pinned wheels have matching verification receipts; targets {"accelerate": "1.15.0", "numpy": "2.5.4", "peft": "0.21.0", "torch": "2.7.1+cu128", "transformers": "5.17.0"}

4. Untouched baseline, training and calibration

The untouched real-model 20-question gate must precede weight updates and bind final manifest/episode hashes. Successful native inference is distinct from mocked tests or CPU head replay. Training proof requires changed tensor hashes, positive weight delta, unchanged released source/adapter and matching reload logits.

Training selects by disjoint tuning-question NLL; separate calibration questions fit each head's temperature using the same predeclared procedure. Neither phase may read fresh outcomes. Better training loss, successful reload and a changed head are not held-out improvement.

Optimizer steps and candidate epochs describe total candidate-fit work across learning-rate trials. The selected saved checkpoint contains its selected trial/epoch prefix; it need not accumulate every candidate's updates.

Proof	Observed
Untouched gate	verified
Attempted / successful / failed / valid forwards	20 / 20 / 0 / 20
Training status / actual head parameters	trained / 3151360
Fit questions / tune questions / rows	60 / 15 / 60+15
Head delta L2 / reload error	0.02861134 / 0.00000000
Selection / elapsed seconds	minimum tuning-question NLL / 0.509
Total candidate-fit epochs / optimizer steps	6 / 12
Selected saved checkpoint LR / epoch	1e-05 / 1
Selected saved checkpoint optimizer steps	2
Source head / adapter unchanged	True / True
Untouched control tensor / adapter linkage	verified
Released head SHA256	ace56742545b505e6fae34ada85c8edbeedb23ba8974d8680aebe0380d6083a1

Proof	Observed
Saved trained head SHA256	d95b640d54ccc004cc136dfbb38d896e942f299d972bed555bf07f7ab648ccf5

Untouched DEVELOPMENT first-draft results

Recomputed from receipt-bound episodes and evaluator tasks, with raw critic responses/public inputs reconciled against the hash-bound native journal. The standalone summary is not used. All original planned questions remain in the denominator; failed episodes score zero.

This is the pre-training cap-one development gate. Fixed1 and completed Strands episodes return the same first answer; there is no revision opportunity, fresh comparison or training gain in these numbers.

Approvals: 11; EM-correct approvals: 5; EM-wrong approvals: 6; rejections of EM-correct drafts: 4. Approval correctness is conditional on approval, not all-question answer accuracy.

Policy	Complete/N	Answer EM	Token F1	Scope
strands-untuned	20/20	45.00%	0.527619	DEVELOPMENT / cap1
fixed-1-prefix	20/20	45.00%	0.527619	DEVELOPMENT / cap1

Calibration evidence

Calibration NLL is measured on the fitting data, not confirmation performance.

Head	Questions	Temperature	NLL T1	NLL fitted
untuned	15	1.438	0.568	0.551
trained	15	1.427	0.567	0.550

Secondary post-training DEVELOPMENT probability diagnostics

The same 20 development questions were inspected before training. These cap1 results describe probabilities on saved first drafts; they establish neither repair benefit, answer improvement nor fresh generalization. Released and trained heads each use their separately fitted CAL temperature; weights and temperatures are not selected again here.

The shared saved writer answers have EM 45.00% and token F1 0.527619. A head failure can reduce its failure-inclusive episode EM, but cannot improve these saved answers. Missing/overflow probabilities remain missing; all 20 stay in coverage denominators. Undefined AUROC or precision is not zero.

Calibrated head	Scored/N	AUROC	Brier	NLL	Precision	Approval/N
untuned calibrated	20/20	0.474747	0.378804	1.000569	0.500000	0.400000
trained calibrated	20/20	0.474747	0.378881	1.000798	0.500000	0.400000

Secondary DEVELOPMENT paired probability differences

Trained minus released calibrated head; 5,000 paired draws, seed 42, original development question as unit. Probability differences use the 20/20 observed pairs; 0 absent pairs are not imputed. Approval coverage uses all 20, with absent decisions counted as no approval. Decision classification accuracy concerns correctness prediction, not answer EM.

AUROC/precision point differences use each arm's observed scores; their paired intervals use only the observed-pair intersection. Nonadditive intervals exclude and count draws undefined in either arm. Additive intervals, when defined, use all 5,000 draws. These small, reused-cohort intervals are descriptive.

Statistic	Point delta	Paired95% interval	Pair N	Valid/undefined draws
nll	0.000229	[0.000074, 0.000391]	20	5,000 / 0
brier	0.000077	[0.000022, 0.000135]	20	5,000 / 0
Decision classification accuracy	0.000000	[0.000000, 0.000000]	20	5,000 / 0
approval_coverage_of_planned	0.000000	[0.000000, 0.000000]	20	5,000 / 0
auroc	0.000000	[0.000000, 0.000000]	20	5000 / 0
approval_precision	0.000000	[0.000000, 0.000000]	20	5000 / 0

5. Matched confirmation quality

All planned questions remain in each denominator. Failed/missing policy episodes score zero for EM/F1. These are terminal common-trajectory replay results; prefix model seconds are not measured live end-to-end latency. Raw reported tokens across different tokenizers are usage records, not equal compute.

Secondary visited-confidence policies: strands-untuned-retainbest, strands-untuned-cap3-retainbest, strands-trained-retainbest, strands-trained-cap3-retainbest. They cannot inspect future drafts, rescue failures or erase prior costs.

Policy	Complete	EM %	F1	Drafts/q	Tokens/q	Prefix s/q
fixed-1-prefix	20/20	50.00	0.5400	1.000	1,917.1	0.5749
fixed-2-prefix	20/20	50.00	0.5400	2.000	3,869.4	0.7447
fixed-3-prefix	20/20	50.00	0.5400	3.000	5,821.7	0.8990
fixed-5-prefix	20/20	50.00	0.5400	5.000	9,726.3	1.2268
fixed-10-prefix	20/20	50.00	0.5400	10.000	19,487.8	2.0599
strands-untuned	20/20	50.00	0.5400	5.500	19,995.0	2.6268
strands-untuned-cap3	20/20	50.00	0.5400	2.000	7,351.1	1.2361
strands-trained	20/20	50.00	0.5400	5.500	19,995.0	2.6267
strands-trained-cap3	20/20	50.00	0.5400	2.000	7,351.1	1.2360

Policy	Complete	EM %	F1	Drafts/q	Tokens/q	Prefix s/q
strands-untuned-retainbest	20/20	50.00	0.5400	5.500	19,995.0	2.6268
strands-untuned-cap3-retainbest	20/20	50.00	0.5400	2.000	7,351.1	1.2361
strands-trained-retainbest	20/20	50.00	0.5400	5.500	19,995.0	2.6267
strands-trained-cap3-retainbest	20/20	50.00	0.5400	2.000	7,351.1	1.2360

Paired differences

Candidate minus baseline, percentage points; 95% paired percentile bootstrap, 5,000 draws, seed 42, original question as sampling unit. Intervals are descriptive and do not establish general superiority, equivalence or non-inferiority. Repeated rounds do not increase independent N.

Contrast	EM delta [95%CI]	F1 delta [95%CI]	Complete pairs
strands-trained_minus_strands-untuned	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained-cap3_minus_strands-untuned-cap3	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned_minus_fixed-1-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned_minus_fixed-10-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned_minus_fixed-2-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned_minus_fixed-3-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned_minus_fixed-5-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained_minus_fixed-1-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained_minus_fixed-10-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained_minus_fixed-2-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained_minus_fixed-3-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained_minus_fixed-5-prefix	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained-retainbest_minus_strands-untuned-retainbest	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained-cap3-retainbest_minus_strands-untuned-cap3-retainbest	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned-retainbest_minus_strands-untuned	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-untuned-cap3-retainbest_minus_strands-untuned-cap3	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20
strands-trained-retainbest_minus_strands-trained	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20

Contrast	EM delta [95%CI]	F1 delta [95%CI]	Complete pairs
strands-trained-cap3-retainbest_minus_strands-trained-cap3	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	20

Probability diagnostics and coverage

AUROC/Brier/NLL use observed first decisions only; missing/context-overflow probabilities remain coverage failures. Null AUROC for one label class is undefined, not zero. Approval precision is conditional on approval; show coverage alongside it.

Policy	Scored/N	AUROC	Brier	NLL	Precision	Approval/N
strands-untuned	20/20	0.380	0.336	0.911	0.500	0.500
strands-untuned-cap3	20/20	0.380	0.336	0.911	0.500	0.500
strands-trained	20/20	0.380	0.336	0.911	0.500	0.500
strands-trained-cap3	20/20	0.380	0.336	0.911	0.500	0.500

Repairs, regressions and approvals

Transitions use answer EM as the recorded correctness proxy. Correct-first-draft loss includes failures; approval precision is conditional and must be read with approval coverage. Missing values are not invented.

Policy	Repairs	Regressions	Correct lost	Wrong approvals	Precision	Approval/N
strands-untuned	0	0	0	5	0.500	0.500
strands-untuned-cap3	0	0	0	5	0.500	0.500
strands-trained	0	0	0	5	0.500	0.500
strands-trained-cap3	0	0	0	5	0.500	0.500
strands-untuned-retainbest	0	0	0	5	0.500	0.500
strands-untuned-cap3-retainbest	0	0	0	5	0.500	0.500
strands-trained-retainbest	0	0	0	5	0.500	0.500
strands-trained-cap3-retainbest	0	0	0	5	0.500	0.500

6. Physical work and interpretation

This table records physical work once per phase; it is not a total system-cost estimate. Shared writer generation and real encoder features are not repeated expenses for each replayed policy. CPU head replay adds zero physical encoder tokens. Training/calibration/startup and intentional idle remain separate evidence; inspect provenance and phase manifests for omitted components.

TRAIN invocation wall=497.867s; intentional idle=450.000s. Resumed invocations must not be confused with cumulative journal usage.

Local hosted API cost of $0 leaves hardware/electricity unpriced. Counterfactual policy prefixes are not additional physical API calls or demonstrated deployment savings. Missing usage stays unknown or a stated lower bound, never fabricated zero. The historical Jev development PDF is a separate cohort/profile and is preserved unchanged.

Phase	Calls/forwards	Input tokens	Output tokens	Model seconds	Hosted USD
TRAIN writer	90	160,957	866	42.251	0.000000
Historical TRAIN preparation	591	1,052,407	5,827	383.055	0.000000
Untouched DEVELOPMENT native decisions	20	34,183	0	24.971	0.000000
Original DEVELOPMENT writer once	20	35,330	212	18.387	0.000000
CALIBRATION untuned CPU head	15	0	0	0.019	0.000000
CALIBRATION trained CPU head	15	0	0	0.015	0.000000
Secondary DEVELOPMENT encoder once	20	34,183	not generative	5.828	0.000000
Secondary DEVELOPMENT untuned CPU head	20	0	0	0.024	0.000000
Secondary DEVELOPMENT trained CPU head	20	0	0	0.020	0.000000
writer_generation_once	200	387,901	1,855	41.198	0.000000
real_encoder_features_once	200	364,275	not generative	47.387	not recorded
untuned_head_scoring	110	0	0	0.115	0.000000
tuned_head_scoring	110	0	0	0.113	0.000000
TRAIN encoder fit	60	103,369	not generative	26.561	not recorded
TRAIN encoder tune	15	25,922	not generative	4.435	not recorded
TRAIN encoder calibration	15	27,409	not generative	4.602	not recorded

Fresh confirmation wall and idle scopes

These are elapsed scopes of the completed physical experiment, not live controller latency. Shared writer generation and native encoding occur once; each CPU replay is a separate physical phase. Request/model seconds in the preceding table are already inside phase wall time and are not added again.

Startup and work-phase wall are separate only where recorded. Startup plus phase is calculated only when both exist; missing startup or partial timing fields remain not recorded. The owning-stage wall also encloses process/wrapper setup and cleanup, so its overlapping wall, request and idle columns must not be summed. A residual is not an exact startup, resource-pause or utilization measurement.

Intentional idle is inside its corresponding startup or phase wall and includes configured rests, holds and scheduling overshoot. Counterfactual policy-prefix seconds remain a separate replay estimate; these physical elapsed times establish no live latency or deployment-saving benefit.

Physical phase	Startup wall s	Startup idle s	Phase wall s	Phase idle s	Startup+phase s	Stage wall s
Writer generation once	not recorded	not recorded	1,051.609	1,000.110	not recorded	1,059.473
Native encoder cache once	14.442	7.203	96.235	45.939	110.677	112.087
Untuned CPU head replay	8.842	5.000	556.071	550.030	564.913	566.269
Trained CPU head replay	8.763	5.000	555.989	550.078	564.752	566.283

Untouched DEVELOPMENT wall and idle scopes

Native request seconds in the physical-work table are a subset of replay-phase wall time. Backend startup is separate from that phase; only startup wall plus phase wall is summed here. Do not add request time again. Writer draft generation is a separate expense and is not included in this native-decision timing.

Phase intentional idle includes rests, resource holds and scheduling overshoot. Their residual is not an exact RAM-pause measurement. Hardware/electricity remain unpriced, and local hosted USD0 is not zero total cost.

Timing scope	Recorded seconds	Interpretation
Backend startup wall	20.751079	Admission/construction/placement/rest
Replay phase wall	270.251957	Native requests, phase rests and holds
Phase intentional idle	241.704000	Inside phase wall; not added again
Startup plus phase wall	291.003036	Separate scopes summed once

Historical TRAIN preparation coverage

This explicitly selected earlier source contributes physical preparation expense only. Its questions, trajectories, references, feature caches and planned denominator are not added to the selected training cohort or confirmation quality. Repeated questions in a new writer run still incur new physical calls.

Historical cumulative journal usage is charged once. Its archived cumulative counters supply timing provenance and are not additional calls/tokens. These rows remain a partial accounting of the campaign, with hardware/electricity and unprovided phases unpriced.

Source state	Planned	Saved	Physical calls	Missing	Completed without saved round
interrupted	600	590	591	10	1

TRAIN invocation timing

Archived invocations are included only when explicitly selected, receipt-hash verified and journal-prefix matched to the current source. Their worker intervals must not overlap. Starts are derived from recorded finish minus measured invocation wall time; source started_at may retain the first invocation timestamp after resume.

Wall/idle sums exclude the gap between invocations, wrapper startup and final unload/cleanup. Physical token/call usage above is already cumulative and is counted once. The first selected snapshot has no proven prior counter; later increments are differences between selected snapshots, not additional expenses to add to the cumulative total. Without selected archives, only the latest invocation is shown.

Invocation	Worker wall s	Intentional idle s	Physical call counter	Recorded finish UTC
Invocation 1 (current)	497.867	450.000	90 cumulative; increment not isolated	2026-10-18T23:33:40.603252+00:00

Historical TRAIN invocation timing

Archived invocations are included only when explicitly selected, receipt-hash verified and journal-prefix matched to the current source. Their worker intervals must not overlap. Starts are derived from recorded finish minus measured invocation wall time; source started_at may retain the first invocation timestamp after resume.

Wall/idle sums exclude the gap between invocations, wrapper startup and final unload/cleanup. Physical token/call usage above is already cumulative and is counted once. The first selected snapshot has no proven prior counter; later increments are differences between selected snapshots, not additional expenses to add to the cumulative total. Without selected archives, only the latest invocation is shown.

Invocation	Worker wall s	Intentional idle s	Physical call counter	Recorded finish UTC
Invocation 1 (archived)	2,224.024	2,000.234	328 cumulative; increment not isolated	2026-10-18T21:15:42.797566+00:00
Invocation 2 (current)	1,527.216	1,314.625	+263 since prior snapshot	2026-10-18T22:15:00.181119+00:00
Sum of selected invocations	3,751.241	3,314.859	591 final cumulative	excludes gaps / unload

7. Two next architectural ideas

Evidence-specialist repair under Strands stop authority. On rejection, a specialist returns short diagnostics cited to supplied evidence before revision; Strands retains final STOP authority, with at most one specialist call per revision and two per question. Pros: inspectable, targeted repairs. Cons: extra cost/latency, wrong diagnostics and more failure modes. Test generic versus cited feedback under matched decisions and count every role's work.

Budget-aware STOP/VERIFY utility learning after the supervised baseline. Use disjoint TRAIN continuations to learn whether verification is worth its token/time cost, initially with bounded one-step full-information targets. Pros: directly aligns decisions with budget and may avoid unnecessary verification. Cons: reward-weight sensitivity, proxy-label errors

and replay/off-policy bias; actual action effects need later matched live validation. Supervised head training is not an RL result, and fresh confirmation must never tune the policy.

Evidence qualifications

Historical TRAIN preparation is a separate physical source: 591 completed calls; 1 completed call(s) without a saved round. Its questions, caches and denominator are not pooled with the selected TRAIN source.

The supplied TRAIN selection manifest preserves its PROPOSED data-preparation label. That label alone establishes neither adoption nor execution; this report's separately verified owning-campaign and training-proof evidence determine what actually ran.

fit features: selected=60; overflow rows=0; unreconciled=0.

tune features: selected=15; overflow rows=0; unreconciled=0.

calibration features: selected=15; overflow rows=0; unreconciled=0.

Evidence references

Input SHA256s and missing-file records are preserved in the provenance JSON. Source paths below identify the actual receipts; they are not model outputs.

[S1] explicit campaign state: runtime/campaign-fresh20-after-v8-2130-v1-20261018/campaign.json (SHA256 399b87fcc071...)

[S2] explicit frozen campaign plan: runtime/campaign-fresh20-after-v8-2130-v1-20261018/plan.json (SHA256 d6b2b3c88f36...)

[S3] separate training frozen plan: runtime/campaign-training-only-train90-v8-20261018/plan.json (SHA256 41cc0f836c76...)

[S4] separate completed training campaign: runtime/campaign-training-only-train90-v8-20261018/campaign.json (SHA256 c4f4bba88cf8...)

[S5] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/calibrate.py (SHA256 12869bbc3da4...)

[S6] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/campaign.py (SHA256 fa928ae64679...)

[S7] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/compare.py (SHA256 d18ad91edbfc...)

[S8] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/control.py (SHA256 bfcc4f506388...)

[S9] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/data.py (SHA256 32d8988a3644...)

[S10] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/decider.py (SHA256 f280e2e8844c...)

[S11] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/evaluate.py (SHA256 f125da37e232...)

[S12] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/jev_replay.py (SHA256 2df3e7f4821b...)

[S13] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/metrics.py (SHA256 1e8d50644e99...)

[S14] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/retention.py (SHA256 c50371fc9309...)

[S15] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/runtime_setup.py (SHA256 4bccf6d50977...)

[S16] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/train_head.py (SHA256 fe1119de024d...)

[S17] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/windows_job.py (SHA256 526fceeaa264...)

[S18] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/writer_service.py (SHA256 9e7f1155bb13...)

[S19] separate campaign frozen runtime: runtime/campaign-training-only-train90-v8-20261018/source/pc_strands/__init__.py (SHA256 87f1356fdf9d...)

[S20] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/calibrate.py (SHA256 12869bbc3da4...)

[S21] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/campaign.py (SHA256 fa928ae64679...)

[S22] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/compare.py (SHA256 d18ad91edbfc...)

[S23] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/control.py (SHA256 bfcc4f506388...)

[S24] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/data.py (SHA256 32d8988a3644...)

[S25] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/decider.py (SHA256 f280e2e8844c...)

[S26] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/evaluate.py (SHA256 f125da37e232...)

[S27] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/jev_replay.py (SHA256 2df3e7f4821b...)

[S28] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/metrics.py (SHA256 1e8d50644e99...)

[S29] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/retention.py (SHA256 c50371fc9309...)

[S30] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/runtime_setup.py (SHA256 4bccf6d50977...)

[S31] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/train_head.py (SHA256 fe1119de024d...)

[S32] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/windows_job.py (SHA256 526fceeaa264...)

[S33] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/writer_service.py (SHA256 9e7f1155bb13...)

[S34] separate campaign frozen runtime: runtime/campaign-fresh20-after-v8-2130-v1-20261018/source/pc_strands/__init__.py (SHA256 87f1356fdf9d...)

[S35] actual training owning-stage output: runtime/study-runs/train90-original-v7-fixed1/manifest.json (SHA256 49c5f38f8d4d...)

[S36] actual training owning-stage output: runtime/study-runs/train90-original-v7-fixed1/trajectories.json (SHA256 6c3ac227dceb...)

[S37] actual training owning-stage output: runtime/study-runs/train90-original-v7-fixed1/requests.jsonl (SHA256 9762b5e4622b...)

[S38] actual training owning-stage output: runtime/study-runs/train90-original-v7-fixed1/tasks-evaluator-only.json (SHA256 e62a9b75e5a0...)

[S39] actual training owning-stage output: runtime/writer-supervision/train90-original-v7-fixed1-ram7-q8-flash/supervisor.json (SHA256 807873f9c30d...)

[S40] actual training owning-stage output: runtime/feature-caches/train90-original-v7/fit.jsonl (SHA256 5114e7a544f5...)

[S41] actual training owning-stage output: runtime/feature-caches/train90-original-v7/fit.jsonl.manifest.json (SHA256 cf16587e72e0...)

[S42] actual training owning-stage output: runtime/feature-caches/train90-original-v7/fit.jsonl.events.jsonl (SHA256 ea9b8971f530...)

[S43] actual training owning-stage output: runtime/feature-caches/train90-original-v7/tune.jsonl (SHA256 2de5a14988e6...)

[S44] actual training owning-stage output: runtime/feature-caches/train90-original-v7/tune.jsonl.manifest.json (SHA256 97e2a5150594...)

[S45] actual training owning-stage output: runtime/feature-caches/train90-original-v7/tune.jsonl.events.jsonl (SHA256 670609c24d7f...)

[S46] actual training owning-stage output: runtime/feature-caches/train90-original-v7/calibration.jsonl (SHA256 426a66fb547a...)

[S47] actual training owning-stage output: runtime/feature-caches/train90-original-v7/calibration.jsonl.manifest.json (SHA256 ef18a9d82e75...)

[S48] actual training owning-stage output: runtime/feature-caches/train90-original-v7/calibration.jsonl.events.jsonl (SHA256 6375498919f3...)

[S49] actual training owning-stage output: runtime/checkpoints/train90-original-v7/head-trained/head.safetensors (SHA256 d95b640d54cc...)

[S50] actual training owning-stage output: runtime/checkpoints/train90-original-v7/head-trained/training-proof.json (SHA256 75c3cbd60aaa...)

[S51] actual training owning-stage output: runtime/checkpoints/train90-original-v7/released-calibrated/head.safetensors (SHA256 2492613e72ea...)

[S52] actual training owning-stage output: runtime/checkpoints/train90-original-v7/released-calibrated/training-proof.json (SHA256 7c5b95e829e3...)

[S53] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/manifest.json (SHA256 f080e9df3c7d...)

[S54] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/episodes.json (SHA256 96f791692f7d...)

[S55] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/summary.json (SHA256 c5dbde01d5a2...)

[S56] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/requests.jsonl (SHA256 84067dc90727...)

[S57] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/tasks-evaluator-only.json (SHA256 d5b5f7a3092d...)

[S58] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/paired-comparisons.json (SHA256 0301ad4bfb03...)

[S59] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/common-schema-provenance.json (SHA256 10c71c411511...)

[S60] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/first-draft-logits.jsonl (SHA256 1d1d177a061b...)

[S61] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-untuned/temperature-fit.json (SHA256 6f96c63df175...)

[S62] actual training owning-stage output: runtime/checkpoints/train90-original-v7/released-calibrated/calibration-proof.json (SHA256 6f96c63df175...)

[S63] actual training owning-stage output: runtime/checkpoints/train90-original-v7/released-calibrated/strands_decider_config.json (SHA256 a75610b67f13...)

[S64] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/manifest.json (SHA256 9f979775f072...)

[S65] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/episodes.json (SHA256 ea325fc6da02...)

[S66] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/summary.json (SHA256 fc114fef6c3a...)

[S67] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/requests.jsonl (SHA256 ee6c747c0bf2...)

[S68] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/tasks-evaluator-only.json (SHA256 d5b5f7a3092d...)

[S69] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/paired-comparisons.json (SHA256 be42b6f96316...)

[S70] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/common-schema-provenance.json (SHA256 1417658db790...)

[S71] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/first-draft-logits.jsonl (SHA256 b67dc917e6cb...)

[S72] actual training owning-stage output: runtime/study-runs/train90-original-v7-calibration-trained/temperature-fit.json (SHA256 4c183ff94383...)

[S73] actual training owning-stage output: runtime/checkpoints/train90-original-v7/head-trained/calibration-proof.json (SHA256 4c183ff94383...)

[S74] actual training owning-stage output: runtime/checkpoints/train90-original-v7/head-trained/strands_decider_config.json (SHA256 c8524a28dd0b...)

[S75] TRAIN resource/control observation: runtime/study-runs/train90-original-v7-fixed1/control-status.json (SHA256 35401665f59a...)

[S76] Historical TRAIN generation manifest: runtime/study-runs/train600-fixed1/manifest.json (SHA256 4656a0060c6e...)

[S77] Historical TRAIN resource/control observation: runtime/study-runs/train600-fixed1/control-status.json (SHA256 106ec970e27b...)

[S78] Historical TRAIN attempted trajectories: runtime/study-runs/train600-fixed1/trajectories.json (SHA256 7161ab2cb6cb...)

[S79] Historical TRAIN original partition metadata: runtime/study-runs/train600-fixed1/tasks-evaluator-only.json (SHA256 a1409aae2697...)

[S80] Historical TRAIN physical request journal: runtime/study-runs/train600-fixed1/requests.jsonl (SHA256 7c790f369ff1...)

[S81] archived TRAIN segment identity: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/snapshot-receipt.json (SHA256 f9f797e3770b...)

[S82] archived TRAIN segment bytes: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/manifest.json (SHA256 748ff15a1666...)

[S83] archived TRAIN segment bytes: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/control-status.json (SHA256 e1b2afec2595...)

[S84] archived TRAIN segment bytes: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/requests.jsonl (SHA256 6f829c493101...)

[S85] archived TRAIN segment bytes: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/trajectories.json (SHA256 67b04875e59e...)

[S86] archived TRAIN segment bytes: runtime/writer-supervision/train600-background-v2-ram7-q8-v5/segment-1-final/tasks-evaluator-only.json (SHA256 a1409aae2697...)

[S87] untouched real-forward gate: runtime/study-runs/untuned-dev20-decisions/untuned-pretraining-receipt.json (SHA256 dbe44ea106b5...)

[S88] released untouched head weights: runtime/models/decider/head.safetensors (SHA256 ace56742545b...)

[S89] untouched receipt bound result: runtime/study-runs/untuned-dev20-decisions/manifest.json (SHA256 e7d09f8ce0ad...)

[S90] untouched receipt bound result: runtime/study-runs/untuned-dev20-decisions/episodes.json (SHA256 f3a9c35f30c1...)

[S91] untouched evaluator task snapshot: runtime/study-runs/untuned-dev20-decisions/tasks-evaluator-only.json (SHA256 645c54385ea1...)

[S92] untouched physical native request journal: runtime/study-runs/untuned-dev20-decisions/requests.jsonl (SHA256 49e520400ef6...)

[S93] released frozen adapter: runtime/models/decider/lora/adapter_model.safetensors (SHA256 ae1d0cd1ac3e...)

[S94] explicit selected TRAIN cohort: project/testing/research-2026-10-18-strands/candidate-train-original90-v7/manifest.json (SHA256 a7c7b0647016...)

[S95] trained frozen adapter identity: runtime/checkpoints/train90-original-v7/head-trained/lora/adapter_model.safetensors (SHA256 ae1d0cd1ac3e...)

[S96] untuned frozen adapter identity: runtime/checkpoints/train90-original-v7/released-calibrated/lora/adapter_model.safetensors (SHA256 ae1d0cd1ac3e...)

[S97] explicit secondary DEVELOPMENT20 diagnostics: runtime/dev20-posttraining-v1-diagnostics.json (SHA256 2dfd902be439...)

[S98] DEVELOPMENT matched training/control/CAL owner: runtime/fresh20-after-v8-training-control-calibration-qualification.json (SHA256 5ecc66f42f14...)

[S99] secondary DEVELOPMENT bound input: runtime/reviewed-campaign-plan-dev20-posttraining-v1.json (SHA256 839833368a52...)

[S100] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/campaign.json (SHA256 ffaf7a62c9a2...)

[S101] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/plan.json (SHA256 839833368a52...)

[S102] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/calibrate.py (SHA256 12869bbc3da4...)

[S103] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/campaign.py (SHA256 fa928ae64679...)

[S104] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/compare.py (SHA256 d18ad91edbfc...)

[S105] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/control.py (SHA256 bfcc4f506388...)

[S106] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/data.py (SHA256 32d8988a3644...)

[S107] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/decider.py (SHA256 f280e2e8844c...)

[S108] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/evaluate.py (SHA256 f125da37e232...)

[S109] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/jev_replay.py (SHA256 2df3e7f4821b...)

[S110] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/metrics.py (SHA256 1e8d50644e99...)

[S111] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/retention.py (SHA256 c50371fc9309...)

[S112] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/runtime_setup.py (SHA256 4bccf6d50977...)

[S113] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/train_head.py (SHA256 fe1119de024d...)

[S114] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/windows_job.py (SHA256 526fceeaa264...)

[S115] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/writer_service.py (SHA256 9e7f1155bb13...)

[S116] secondary DEVELOPMENT bound input: runtime/campaign-dev20-posttraining-v1-20261018/source/pc_strands/__init__.py (SHA256 87f1356fdf9d...)

[S117] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/SOURCE-FREEZE.json (SHA256 b9ad063d1a2f...)

[S118] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/calibrate.py (SHA256 12869bbc3da4...)

[S119] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/campaign.py (SHA256 fa928ae64679...)

[S120] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/compare.py (SHA256 d18ad91edbfc...)

[S121] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/control.py (SHA256 bfcc4f506388...)

[S122] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/data.py (SHA256 32d8988a3644...)

[S123] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/decider.py (SHA256 f280e2e8844c...)

[S124] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/evaluate.py (SHA256 f125da37e232...)

[S125] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/jev_replay.py (SHA256 2df3e7f4821b...)

[S126] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/metrics.py (SHA256 1e8d50644e99...)

[S127] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/retention.py (SHA256 c50371fc9309...)

[S128] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/runtime_setup.py (SHA256 4bccf6d50977...)

[S129] secondary DEVELOPMENT bound input:
project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/train_head.py (SHA256 fe1119de024d...)

[S130] secondary DEVELOPMENT bound input:
project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/windows_job.py (SHA256 526fceeaa264...)

[S131] secondary DEVELOPMENT bound input:
project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/writer_service.py (SHA256 9e7f1155bb13...)

[S132] secondary DEVELOPMENT bound input:
project/testing/research-2026-10-18-strands/candidate-dev20-posttraining-v1/source/pc_strands/__init__.py (SHA256 87f1356fdf9d...)

[S133] secondary DEVELOPMENT bound input: runtime/campaign-main-v3-baseline-20261018/plan.json (SHA256 7bd5da1ac7a1...)

[S134] secondary DEVELOPMENT bound input: project/presentation/build/evidence-yash-8736dd9/data/hotpotqa/tasks.jsonl (SHA256 9bcdd373cb93...)

[S135] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-fixed1-v3/manifest.json (SHA256 b4201ecde649...)

[S136] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-fixed1-v3/trajectories.json (SHA256 96eca8606a1e...)

[S137] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-fixed1-v3/requests.jsonl (SHA256 0c4c04083ecd...)

[S138] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-fixed1-v3/tasks-evaluator-only.json (SHA256 645c54385ea1...)

[S139] secondary DEVELOPMENT bound input: runtime/reviewed-campaign-plan-v8-training-only-train90.json (SHA256 41cc0f836c76...)

[S140] secondary DEVELOPMENT bound input: runtime/v8-terminal-training-calibration-independent-audit-20261018.json (SHA256 70fc5c073ff5...)

[S141] secondary DEVELOPMENT bound input: runtime/download-status.json (SHA256 1abf3c92cbb7...)

[S142] secondary DEVELOPMENT bound input: runtime/install-status.json (SHA256 15bb329eb572...)

[S143] secondary DEVELOPMENT bound input: runtime/health.json (SHA256 da0c4d2dcb9a...)

[S144] secondary DEVELOPMENT bound input: runtime/models/base/LICENSE (SHA256 50cbab8a892c...)

[S145] secondary DEVELOPMENT bound input: runtime/models/base/config.json (SHA256 ed1c1723241f...)

[S146] secondary DEVELOPMENT bound input: runtime/models/base/merges.txt (SHA256 a9d356d7bdf1...)

[S147] secondary DEVELOPMENT bound input: runtime/models/base/model.safetensors-00001-of-00001.safetensors (SHA256 928acbf11878...)

[S148] secondary DEVELOPMENT bound input: runtime/models/base/model.safetensors.index.json (SHA256 74d2ddfe79f1...)

[S149] secondary DEVELOPMENT bound input: runtime/models/base/preprocessor_config.json (SHA256 27225450ac9c...)

[S150] secondary DEVELOPMENT bound input: runtime/models/base/tokenizer.json (SHA256 fe000e3ed39e...)

[S151] secondary DEVELOPMENT bound input: runtime/models/base/tokenizer_config.json (SHA256 e611fbccc7c2...)

[S152] secondary DEVELOPMENT bound input: runtime/models/base/video_preprocessor_config.json (SHA256 d039cd7d88b3...)

[S153] secondary DEVELOPMENT bound input: runtime/models/base/vocab.json (SHA256 ce99b4cb2983...)

[S154] secondary DEVELOPMENT bound input: runtime/models/decider/LICENSE.md (SHA256 c0d777a0fed0...)

[S155] secondary DEVELOPMENT bound input: runtime/models/decider/chat_template.jinja (SHA256 273d8e0e683b...)

[S156] secondary DEVELOPMENT bound input: runtime/models/decider/lora/adapter_config.json (SHA256 0ced130cefba...)

[S157] secondary DEVELOPMENT bound input: runtime/models/decider/provenance.json (SHA256 8713fedb4e67...)

[S158] secondary DEVELOPMENT bound input: runtime/models/decider/strands_decider_config.json (SHA256 1d9625055e88...)

[S159] secondary DEVELOPMENT bound input: runtime/models/decider/tokenizer.json (SHA256 a2cdd2e10856...)

[S160] secondary DEVELOPMENT bound input: runtime/models/decider/tokenizer_config.json (SHA256 8671bed7c852...)

[S161] secondary DEVELOPMENT bound input: runtime/models/decider/train_config.json (SHA256 85e5dbda4fb6...)

[S162] secondary DEVELOPMENT bound input: runtime/dev20-posttraining-v1-operator-freeze.json (SHA256 13e97884bca9...)

[S163] secondary DEVELOPMENT bound input: runtime/feature-caches/dev20-posttraining-v1/firstdrafts.jsonl (SHA256 16a48b4e5e35...)

[S164] secondary DEVELOPMENT bound input: runtime/feature-caches/dev20-posttraining-v1/firstdrafts.jsonl.manifest.json (SHA256 d1865f43a573...)

[S165] secondary DEVELOPMENT bound input: runtime/feature-caches/dev20-posttraining-v1/firstdrafts.jsonl.events.jsonl (SHA256 bbc05f9aa8ee...)

[S166] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/manifest.json (SHA256 db43f73a76b5...)

[S167] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/episodes.json (SHA256 8863f09877c1...)

[S168] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/requests.jsonl (SHA256 48ea7a385764...)

[S169] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/tasks-evaluator-only.json (SHA256 645c54385ea1...)

[S170] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/summary.json (SHA256 0e44930971ca...)

[S171] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/first-draft-logits.jsonl (SHA256 c64662940b0c...)

[S172] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/common-schema-provenance.json (SHA256 1ee14900a192...)

[S173] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/manifest.json (SHA256 636fc8f87e87...)

[S174] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/episodes.json (SHA256 49e15e676604...)

[S175] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/requests.jsonl (SHA256 4a395937792e...)

[S176] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/tasks-evaluator-only.json (SHA256 645c54385ea1...)

[S177] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/summary.json (SHA256 803d49952880...)

[S178] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/first-draft-logits.jsonl (SHA256 b16e66811480...)

[S179] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/common-schema-provenance.json (SHA256 2f60d5cf143b...)

[S180] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/CANDIDATE-FRESH20-ONLY-AFTER-V8-20261018.json (SHA256 f55472c8aaad...)

[S181] secondary DEVELOPMENT bound input: runtime/training-only-train90-v8-cache-qualification.json (SHA256 8415887e4b2c...)

[S182] secondary DEVELOPMENT bound input: runtime/training-only-train90-v8-operator-freeze.json (SHA256 8f0007ca6c02...)

[S183] secondary DEVELOPMENT bound input: runtime/fresh-freeze-history/fresh-N-freeze-v5-before-v8-training.json (SHA256 27f6d64c89ff...)

[S184] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-train-original90-v7/tasks.jsonl (SHA256 1a789919c3d5...)

[S185] secondary DEVELOPMENT bound input: project/testing/research-2026-10-18-strands/candidate-train-original90-v7/CANDIDATE-VERIFICATION.json (SHA256 d56b079c0fd3...)

[S186] secondary DEVELOPMENT bound input: external/python.exe (SHA256 4d6f5f81a4bc...)

[S187] secondary DEVELOPMENT bound input: project/runtime/ollama-v0.32.15/ollama.exe (SHA256 0a9d42eabc59...)

[S188] secondary DEVELOPMENT bound input: runtime/campaign-main-v3-baseline-20261018/campaign.json (SHA256 3f9509579b75...)

[S189] secondary DEVELOPMENT bound input: runtime/study-runs/train90-original-v7-fixed1/source_snapshot/control.py (SHA256 bfcc4f506388...)

[S190] secondary DEVELOPMENT bound input: runtime/study-runs/train90-original-v7-fixed1/source_snapshot/data.py (SHA256 32d8988a3644...)

[S191] secondary DEVELOPMENT bound input: runtime/study-runs/train90-original-v7-fixed1/source_snapshot/evaluate.py (SHA256 f125da37e232...)

[S192] secondary DEVELOPMENT bound input: runtime/study-runs/train90-original-v7-fixed1/source_snapshot/metrics.py (SHA256 1e8d50644e99...)

[S193] secondary DEVELOPMENT bound input: project/testing/preflight-2026-10-18/hotpot_dev_distractor_v1.hf-rows-verified.json (SHA256 674e6c8b4e8c...)

[S194] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/released-calibrated/chat_template.jinja (SHA256 273d8e0e683b...)

[S195] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/released-calibrated/lora/adapter_config.json (SHA256 0ced130cefba...)

[S196] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/released-calibrated/tokenizer.json (SHA256 a2cdd2e10856...)

[S197] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/released-calibrated/tokenizer_config.json (SHA256 8671bed7c852...)

[S198] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/head-trained/chat_template.jinja (SHA256 273d8e0e683b...)

[S199] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/head-trained/lora/adapter_config.json (SHA256 0ced130cefba...)

[S200] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/head-trained/tokenizer.json (SHA256 a2cdd2e10856...)

[S201] secondary DEVELOPMENT bound input: runtime/checkpoints/train90-original-v7/head-trained/tokenizer_config.json (SHA256 8671bed7c852...)

[S202] secondary DEVELOPMENT bound input: runtime/candidate-panels/fresh20/tasks.jsonl (SHA256 6584e2bf7a66...)

[S203] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/source_snapshot/control.py (SHA256 bfcc4f506388...)

[S204] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/source_snapshot/data.py (SHA256 32d8988a3644...)

[S205] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/source_snapshot/evaluate.py (SHA256 f125da37e232...)

[S206] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-untuned-calibrated/source_snapshot/metrics.py (SHA256 1e8d50644e99...)

[S207] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/source_snapshot/control.py (SHA256 bfcc4f506388...)

[S208] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/source_snapshot/data.py (SHA256 32d8988a3644...)

[S209] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/source_snapshot/evaluate.py (SHA256 f125da37e232...)

[S210] secondary DEVELOPMENT bound input: runtime/study-runs/dev20-posttraining-v1-trained-calibrated/source_snapshot/metrics.py (SHA256 1e8d50644e99...)

[S211] pinned read-only DEVELOPMENT auditor: runtime/operator_dev20_diagnostics.py (SHA256 5fa27dcaa090...)

[S212] pinned read-only DEVELOPMENT auditor: runtime/operatorqualification.py (SHA256 2ecdc14f6778...)

[S213] answer-blind fresh cohort freeze: runtime/fresh-N-freeze.json (SHA256 99f933742e98...)

[S214] campaign copied pre-score freeze: runtime/campaign-fresh20-after-v8-2130-v1-20261018/fresh-freeze.json (SHA256 99f933742e98...)

[S215] fresh generation start and cleanup: runtime/study-runs/fresh20-2130-v1-fixed10/manifest.json (SHA256 8f829d8d2a89...)

[S216] terminal paired confirmation analysis: runtime/study-runs/fresh20-2130-v1-paired-analysis/paired-analysis.json (SHA256 9a4e5114435c...)

[S217] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/manifest.json (SHA256 47cc1013840f...)

[S218] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/episodes.json (SHA256 cdc5cedc62da...)

[S219] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/requests.jsonl (SHA256 0127da08b480...)

[S220] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/tasks-evaluator-only.json (SHA256 ad151baaadbd...)

[S221] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/manifest.json (SHA256 a7236e98f93d...)

[S222] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/episodes.json (SHA256 f081f28eda40...)

[S223] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/requests.jsonl (SHA256 0c32022ae746...)

[S224] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/tasks-evaluator-only.json (SHA256 ad151baaadbd...)

[S225] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-fixed10/trajectories.json (SHA256 c426abe1be2d...)

[S226] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-fixed10/tasks-evaluator-only.json (SHA256 ad151baaadbd...)

[S227] comparison-bound source input: runtime/feature-caches/fresh20-2130-v1-allrounds.jsonl (SHA256 576ad6de6dad...)

[S228] comparison-bound source input: runtime/feature-caches/fresh20-2130-v1-allrounds.jsonl.manifest.json (SHA256 f0998437e52d...)

[S229] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/source_snapshot/control.py (SHA256 bfcc4f506388...)

[S230] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/source_snapshot/data.py (SHA256 32d8988a3644...)

[S231] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/source_snapshot/evaluate.py (SHA256 f125da37e232...)

[S232] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-untuned/source_snapshot/metrics.py (SHA256 1e8d50644e99...)

[S233] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/source_snapshot/control.py (SHA256 bfcc4f506388...)

[S234] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/source_snapshot/data.py (SHA256 32d8988a3644...)

[S235] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/source_snapshot/evaluate.py (SHA256 f125da37e232...)

[S236] comparison-bound source input: runtime/study-runs/fresh20-2130-v1-trained/source_snapshot/metrics.py (SHA256 1e8d50644e99...)

[S237] comparison-bound source input: runtime/feature-caches/fresh20-2130-v1-allrounds.jsonl.events.jsonl (SHA256 3a29a3b8cad6...)

[S238] independently rejoined canonical input: project/testing/strands-work/pc_strands/compare.py (SHA256 d18ad91edbfc...)

[S239] independently rejoined canonical input: project/testing/strands-work/pc_strands/metrics.py (SHA256 1e8d50644e99...)

[S240] independently rejoined canonical input: project/testing/strands-work/pc_strands/retention.py (SHA256 c50371fc9309...)

[S241] official TRAIN selection and exclusion audit: project/testing/research-2026-10-18-strands/official-train600-candidate/manifest.json (SHA256 76f1c0ff4402...)

[S242] historical methods and HotpotQA attribution: project/testing/preflight-2026-10-18/PROFESSOR-METHODS-DRAFT.txt (SHA256 d6be6a5210c9...)

[S243] actual package versions: runtime/installed-packages.json (SHA256 23b5ce1e9017...)

[S244] pinned wheel targets: runtime/dependency-lock.json (SHA256 ff4f81276f48...)

[S245] public pinned download queue: runtime/download-queue.json (SHA256 31fed1b9b6a9...)

[S246] download verification receipts: runtime/download-receipts.json (SHA256 9de87237d5c1...)

[S247] dated hardware inventory: project/pc-evidence/hardware.json (SHA256 e3dc627044aa...)

[S248] captured physical-phase input: runtime/writer-supervision/campaign-dev20-fixed1-v3-ram7-q8-flash/supervisor.json (SHA256 d28058a8f99b...)